

Customer Shopping Behavior Analysis

Uncovering Spending Patterns, Segmentation & Strategic Insights

3,900

Transactions

18

Features

4

Categories

50

US States

Business Problem & Objectives

Business Problem

A retail company observed shifts in customer purchasing patterns across demographics, product categories, and channels.

Key Question:

"How can shopping data be leveraged to improve customer engagement and optimize marketing & product strategies?"

Project Objectives

1. Analyze spending patterns by gender, age & season
2. Identify high-value products & top-rated items
3. Compare subscriber vs non-subscriber behavior
4. Segment customers: New, Returning & Loyal
5. Evaluate impact of discounts on revenue
6. Rank top products within each category

Dataset Overview



3,900

Total Records



18

Columns



4

Categories



37

Null Values
(Review Rating)

Key Features

Feature Group	Columns
Demographics	Age, Gender, Location
Purchase Details	Item, Category, Amount, Season, Size, Color
Behavior	Discount, Frequency, Previous Purchases, Review Rating
Payment / Shipping	Payment Method, Shipping Type, Subscription Status
Engineered	Age Group, Purchase Frequency Days

Data Cleaning & Feature Engineering

01

Null Handling

37 nulls in Review Rating filled using category-level mean imputation via `groupby` transform

02

Column Standardization

Column names lowercased, spaces → underscores, `purchase_amount_(usd)` renamed to `purchase_amount`

03

Age Group Column

Customers segmented into Young Adult, Adult, Middle-aged, Senior using `pd.qcut` (quartile-based)

04

Purchase Frequency Days

Mapped text frequencies (Weekly, Monthly...) to numeric days for quantitative analysis

05

Duplicate Column Drop

`promo_code_used` found 100% identical to `discount_applied` — dropped to eliminate redundancy

06

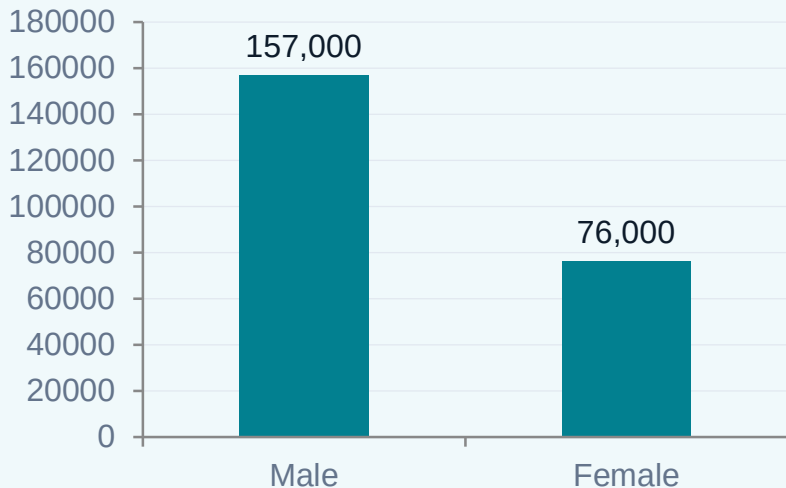
ETL to PostgreSQL

Cleaned DataFrame loaded into PostgreSQL via SQLAlchemy (`psycopg2`) for SQL-based analysis

SQL Insights (1-2): Revenue by Gender & Discount Buyers

Q1 · Revenue by Gender

```
select gender, sum(purchase_amount) as revenue
from customer group by gender
```



Q2 · Discount Buyers > Avg Spend

```
select customer_id, purchase_amount
from customer
where discount_applied='Yes'
and purchase_amount >= (select avg(purchase_amount))
```

\$59.76

Average Purchase

~1,850

Qualifying Customers

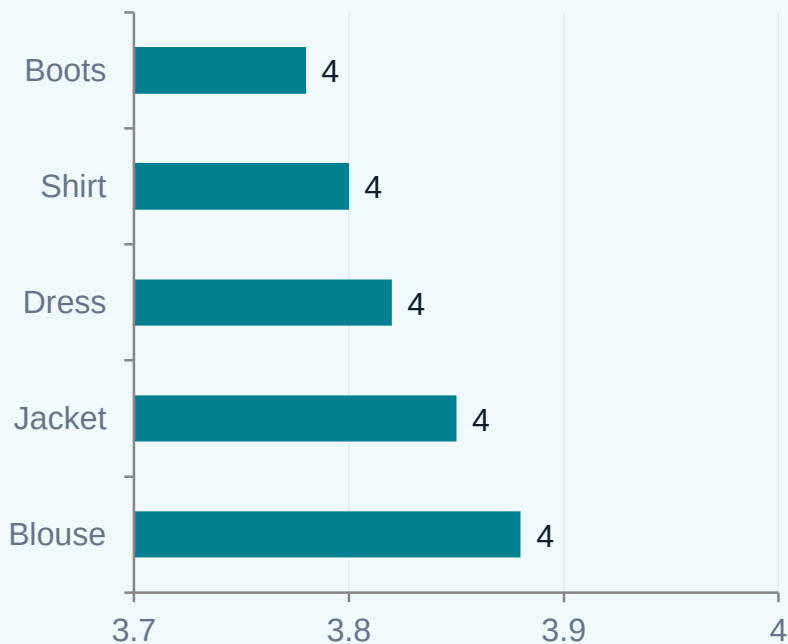
Yes

Discounts Drive High Spend

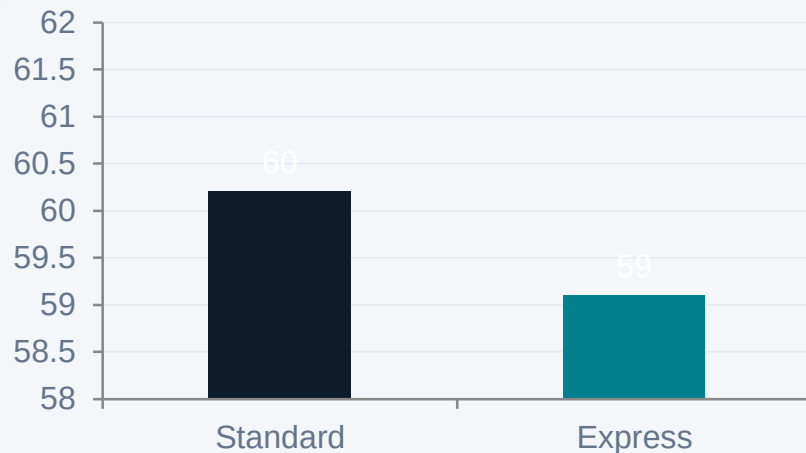
💡 *Discount buyers who still exceed average spend are high-value customers — strong candidates for loyalty programs.*

SQL Insights (3-4): Product Ratings & Shipping Impact

Q3 · Top 5 Products by Avg Review Rating



Q4 · Standard vs Express Shipping

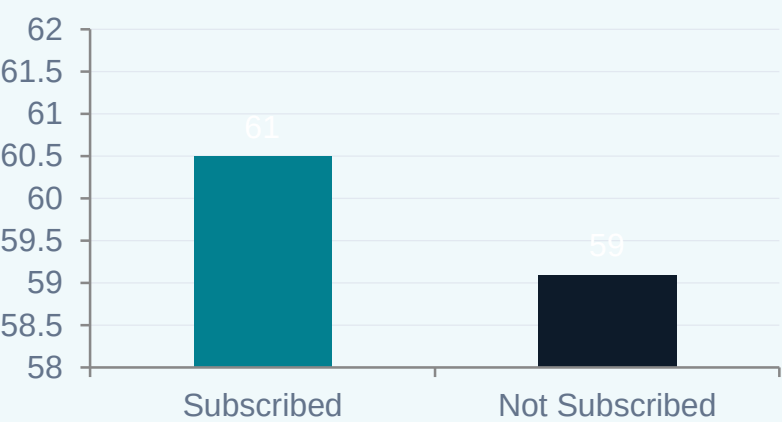


! Shipping type has minimal impact on purchase amount — customer spending behavior is independent of delivery preference.

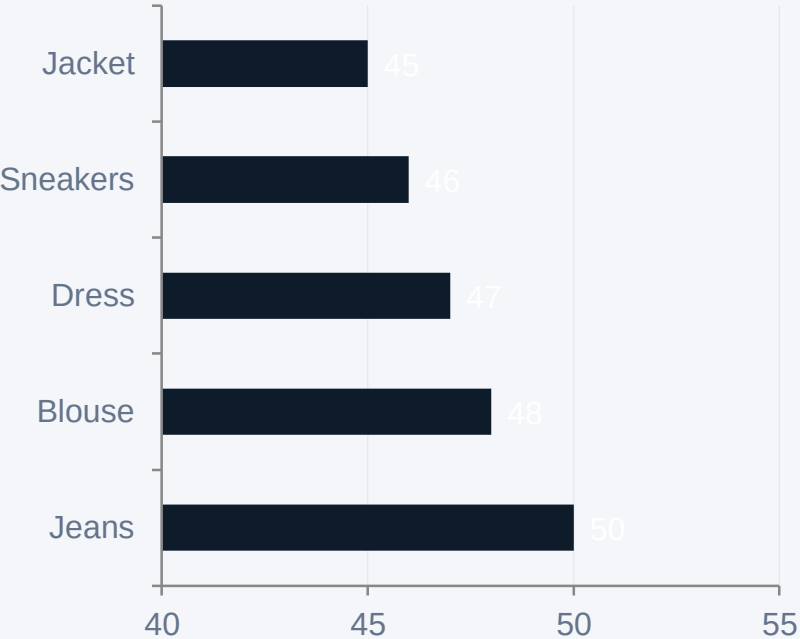
SQL Insights (5-6): Subscriptions & Discount Rates

Q5 · Subscriber vs Non-Subscriber Spend

Group	Customers	Avg Spend	Revenue
Subscribed	1,860	\$60.5	\$112,530
Not Subscribed	2,040	\$59.1	\$120,564



Q6 · Top 5 Products by Discount Rate

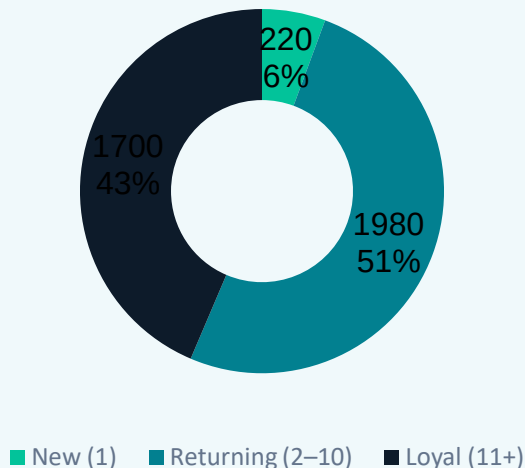


SQL Insights (7 & 9): Customer Segmentation

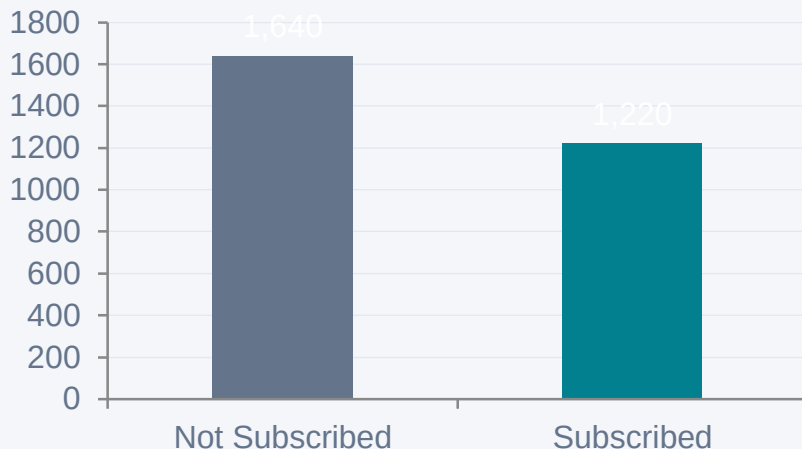
Q7 · Customer Segments (CTE-Based)

Classified using CASE WHEN on previous_purchases:

- New = 1 purchase
- Returning = 2–10
- Loyal = 11+



Q9 · Repeat Buyers (>5 Purchases) & Subscription



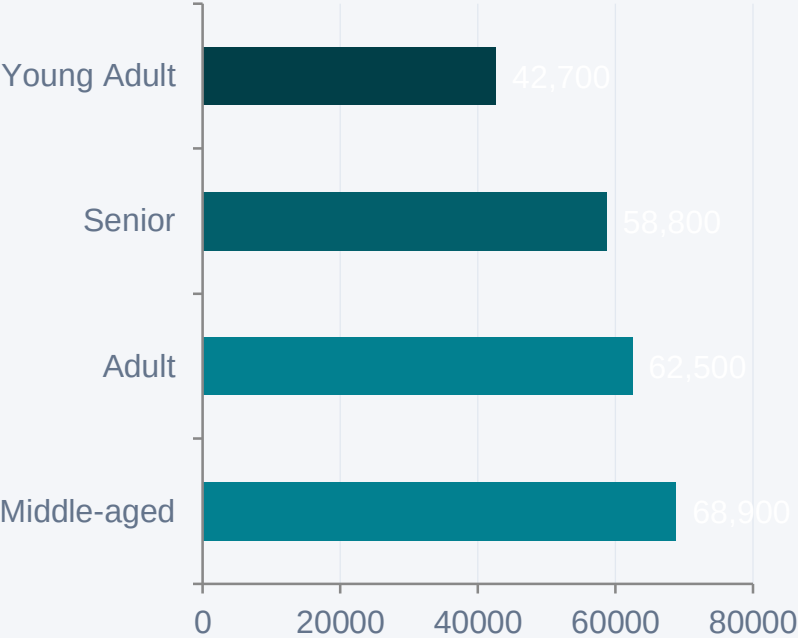
! Repeat buyers are more likely to be non-subscribers — indicating an opportunity to convert loyal customers into subscribers.

SQL Insights (8 & 10): Category Ranking & Age Revenue

Q8 · Top 3 Products Per Category (Window Function)

Rank	Category	Product	Orders
1	Clothing	Blouse	171
2	Clothing	T-Shirt	163
3	Clothing	Shirt	152
1	Accessories	Jewelry	95
2	Accessories	Sunglasses	91
3	Accessories	Belt	88
1	Footwear	Sandals	86
2	Footwear	Sneakers	84
3	Footwear	Boots	80
1	Outerwear	Coat	70
2	Outerwear	Jacket	68
3	Outerwear	Hoodie	65

Q10 · Revenue by Age Group



Key Insights & Takeaways

Actionable findings from 3,900 customer transactions



Male customers drive 67% of total revenue

Targeted campaigns for female segments can improve balance



Loyal customers are largely non-subscribers

Retention offer to convert repeat buyers into subscribers



Discount buyers still spend above average

Strong candidates for loyalty & premium subscription upsell



Clothing dominates all categories

Blouse & T-Shirt lead — prioritize inventory & promotions here



Middle-aged customers are top revenue contributors

Focus marketing budgets on the 45–57 age bracket



Product ratings are consistent (3.78–3.88)

Marginal rating gap means quality improvement has low ROI